

Classification of Delta Band Motor Imagery EEG Signals in SCI Patients using the Regularized Common Temporal Pattern Method

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Abstract—Accurate classification of motor imagery tasks is essential for enhancing brain–computer interfaces (BCIs) in spinal cord injury (SCI) rehabilitation. The use of raw time samples often overlooks temporal dependencies in electroencephalography (EEG), limiting classification accuracy. We propose the Regularized Common Temporal Pattern (RCTempP), a novel time-domain feature extraction method that emphasizes discriminative temporal samples. RCTempP was evaluated on EEG recordings from SCI patients imagining five distinct hand movements within the 0.3–3 Hz band. Compared to raw time samples, Common Temporal Pattern (CTP), and Extended CTP (ECTP) approaches, RCTempP yielded statistically significant improvements in eight out of ten class pairs, for which the average accuracy across subjects ranged from 70.4% to 75.1%. Performance was assessed using a fivefold cross-validation protocol to ensure robust and generalizable results. Importantly, RCTempP's significant temporal filters emerged post-cue onset and corresponded to movement-related cortical potential peaks. These findings highlight RCTempP's promise for advancing motor imagery BCIs in SCI rehabilitation.

Keywords—Brain Computer Interface, Spinal Cord Injury, Motor Imagery, Feature Extraction, Regularized Common Temporal Pattern

I. INTRODUCTION

Brain-computer interfaces (BCIs), particularly when integrated with upper-limb neuroprostheses, hold significant promise for restoring motor function and promoting rehabilitation in individuals with spinal cord injury (SCI) [1], [2], [3], [4]. Among the primary modalities for recording brain activity in BCIs—electroencephalography (EEG), electrocorticography (ECoG), and magnetoencephalography (MEG)—EEG is widely favored due to its non-invasiveness, affordability, and sufficient temporal resolution. Consequently, EEG-based BCIs are commonly used in SCI rehabilitation, especially for motor imagery (MI) tasks involving hand movements. In these tasks, general motor states can be inferred from higher-frequency EEG components, such as mu and beta band power [5], [6], [7], [8], while finer movement parameters—such as speed, force, direction, and grip type—are more effectively decoded from lower-frequency EEG and ECoG signals [9], [10], [11], [12], [13]. These findings have prompted several studies to emphasize the importance of low-frequency bands—particularly the delta band—for extracting more detailed and task-relevant motor information, as movement-related cortical potentials (MRCPs) primarily reside in lower frequency ranges [14], [15]. As such, improving the classification of

hand movement signals remains a central challenge in the development of MI-based BCIs for SCI patients. While end-to-end deep learning approaches have recently shown strong performance in EEG classification tasks [16], simpler models coupled with explicit feature extraction techniques continue to be widely used due to their lower computational complexity and enhanced neuroscientific interpretability. Among these, the Common Spatial Pattern (CSP) algorithm remains one of the most prevalent methods for MI classification [17]. However, CSP relies on either broad or manually selected frequency bands, which can limit its effectiveness [18], [19]. To mitigate this, Filter Bank CSP (FBCSP) was introduced to automatically identify relevant frequency bands [20], yet its performance drops in the low-frequency domain, which is especially relevant to detailed motor decoding [10], [12]. This limitation arises in part because movement-related EEG activity originates from multiple regions—including the primary motor cortex and supplementary motor area—that often produce overlapping spatial patterns [21], [22]. As a result, spatial information alone may be insufficient to distinguish complex limb movements. To address this, advanced techniques have integrated spatial and frequency information, such as ERD/ERS frameworks leveraging Riemannian geometry and partial least squares, which have demonstrated improved performance over traditional CSP-based methods [23]. Building upon these advances, recent research has increasingly focused on combining spatial, temporal, and non-linear features to better represent the complex dynamics of EEG signals. For example, hybrid approaches using Variational Mode Decomposition (VMD), Sample Entropy (SAMPEN), and CSP have yielded significant improvements in classification accuracy [24]. Additionally, graph-theoretical metrics like nodal degree and clustering coefficients have been used to decode movement direction [25], while traditional statistical features from the beta band continue to prove useful for distinguishing movement types [26]. Among these approaches, particular attention has been paid to low-frequency time-domain EEG samples, which have demonstrated strong discriminatory power in classifying various types of hand and arm movements in both healthy individuals and SCI patients [15], [22]. When combined with methods such as discriminative spatial pattern analysis and linear discriminant analysis (LDA), time-domain features have proven especially valuable. However, existing time-domain approaches often rely on raw samples, without leveraging temporal structures or assigning relative importance to specific time points. This highlights the need for novel methods that can extract richer, more meaningful temporal information from EEG signals.

Another study involves the use of connectivity-based features, particularly those derived from multivariate autoregressive (MVAR) models. These models have revealed dynamic patterns of information flow during MI tasks in SCI patients. Notably, the somatosensory cortex dominates information flow prior to cue onset (within alpha and beta bands), while the primary motor cortex becomes more prominent post-cue (mainly in the delta band) [27]. These findings also underscore the importance of temporal dynamics in MI decoding and support the use of central EEG channels, such as Cz, for identifying informative time segments [15].

Despite these insights, time-domain feature extraction methods that move beyond raw sample analysis remain relatively underexplored. The Regularized Common Tensor Pattern (RCTP) algorithm was introduced to address this gap by extending the Common Temporal Pattern (CTP) method to incorporate structured a priori information across time, frequency, and spatial dimensions [28]. Originally applied in a Rapid Serial Visual Presentation (RSVP) speller paradigm, RCTP was shown to outperform standard methods such as CSP and bilinear CSP by assigning greater weights to discriminative time samples. A related method, Extended Common Spatio-Temporal Pattern (ECSTP), further improved classification by combining temporal and spatial features using class-wise trial averages—an approach particularly effective in paradigms like P300 [29], [30].

In this study, we propose a method to improve class discriminability in motor imagery EEG by transforming signals into a subspace that emphasizes informative time samples. Derived from the RCTP framework and restricted to the time domain, this method introduces a data-driven regularization term based on R-squared values from training data—enhancing temporally discriminative information without incorporating information from the spatial or frequency domains. To focus on movement-related cortical potentials (MRCPs), the analysis is limited to the delta band (0.3–3 Hz), which captures the most relevant low-frequency activity. We assess the method's performance by comparing its classification accuracy against raw time samples and existing temporal techniques, including CTP and Extended Common Temporal Pattern (ECTP). Additionally, we validate the learned temporal filters using statistical tests to confirm their alignment with periods of maximal class separability—particularly in the Cz channel. These filters also reveal subject- and task-specific temporal patterns that may inform future BCI design.

This paper is organized as follows. Section II introduces the dataset, preprocessing, and the feature extraction and classification methods. Section III presents the experimental results. Section IV discusses the implications of these findings and their relevance to BCI performance and SCI rehabilitation. Finally, Section V concludes the work and outlines future directions.

II. METHODOLOGY

A. Dataset Description

In this study, we use the EEG dataset from [15], recorded from 10 individuals with cervical spinal cord injury as they attempted five hand movements: Supination (SN), Pronation (PN), Hand Open (HO), Palmar Grasp (PG), and Lateral Grasp (LG). EEG signals (61 channels) and EOG signals (3 channels) were sampled at 256 Hz and appropriately filtered. Each subject completed 9 runs of 40 trials, yielding 72 trials

per movement class. Five subjects (P01, P03, P06, P07, P08) were randomly selected for analysis to assess the performance of the proposed method. Although this subset includes data of varying quality, it serves to evaluate the feasibility and effectiveness of our approach.

B. Preprocessing Procedure

The preprocessing procedure applied to the dataset was partly inspired by the approach described in [15], but several key differences were introduced. Importantly, all preprocessing steps in this study were implemented independently, without relying on external toolboxes. First, the AFz channel was excluded. The signals were then filtered using a 4th-order zero-phase Butterworth bandpass filter between 0.3 Hz and 70 Hz to remove noise and slow drifts. Next, Principal Component Analysis (PCA) was applied to 20-second non-overlapping windows, retaining 99% of the signal variance. Within each window, Independent Component Analysis (ICA) weight matrices were computed and used to extract Independent Components (ICs). Artifact-contaminated ICs—primarily those reflecting eye or muscle activity—were visually identified and removed. The remaining ICs were projected back into the time domain, after which the 20-second windows were concatenated. The signals were then filtered again in the 0.3–3 Hz range to isolate low-frequency oscillations. Finally, trials were extracted and downsampled to 16 Hz, approximately three times the Nyquist rate, to preserve signal content up to 3 Hz.

C. Common Temporal Pattern (CTP)

The CSP method computes a spatial filter matrix that transforms the original time-domain signal into a new space where the variance of the resulting signals is optimized to maximize class separability [17]. Similarly, the CTP method applies the same principle using a temporal filter matrix instead of a spatial one [31]. In this method, the observation matrix is represented as $\mathbf{X} \in \mathbb{R}^{T \times N}$, where T represents the number of temporal samples, and N represents the number of channels. The observation matrix, $\mathbf{X}$, is re-referenced using the Common Average Reference (CAR) method. In the following sections, we assume two classes, with M_k trials in class k (for $k = 1, 2$), and denote the i -th trial of class k by $\mathbf{X}_k^{(i)} \in \mathbb{R}^{T \times N}$. The CTP temporal filters are obtained by extremizing the following objective function:

$$J_{CTP}(\mathbf{w}) = \frac{\mathbf{w}^T \left(\sum_{i=1}^{M_1} \mathbf{X}_1^{(i)} \mathbf{X}_1^{(i)T} \right) \mathbf{w}}{\mathbf{w}^T \left(\sum_{i=1}^{M_2} \mathbf{X}_2^{(i)} \mathbf{X}_2^{(i)T} \right) \mathbf{w}} = \frac{\mathbf{w}^T \mathbf{C}_1 \mathbf{w}}{\mathbf{w}^T \mathbf{C}_2 \mathbf{w}} \quad (1)$$

where $\mathbf{C}_1$ and $\mathbf{C}_2$ are estimations of the covariance matrices of the first and second classes, respectively. The optimization problem can be solved using the Lagrange method:

$$\mathcal{L}(\lambda, \mathbf{w}) = \mathbf{w}^T \mathbf{C}_1 \mathbf{w} - \lambda (\mathbf{w}^T \mathbf{C}_2 \mathbf{w} - 1) \quad (2)$$

where $\mathcal{L}$ is the Lagrangian function and λ is the Lagrange multiplier. Thus, we have:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2\mathbf{w}^T \mathbf{C}_1 - 2\lambda \mathbf{w}^T \mathbf{C}_2 \stackrel{\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 0}{\implies} \mathbf{C}_1 \mathbf{w} = \lambda \mathbf{C}_2 \mathbf{w} \quad (3)$$

$$\stackrel{\det(\mathbf{C}_2) \neq 0}{\implies} (\mathbf{C}_2^{-1} \mathbf{C}_1) \mathbf{w} = \lambda \mathbf{w}$$

Accordingly, the eigenvalues and eigenvectors of $\mathbf{C}_2^{-1}\mathbf{C}_1$ yield λ and $\mathbf{w}$, respectively. The CTP matrix is constructed by concatenating the eigenvectors column-wise:

$$\mathbf{W}_{CTP} = [\mathbf{v}_1, \dots, \mathbf{v}_F, \mathbf{v}_{T-F+1}, \dots, \mathbf{v}_T] \in \mathbb{R}^{T \times 2F} \quad (4)$$

where $\mathbf{v}_m$ represents the m -th eigenvector, and $2F$ is the number of temporal filters. To transform the re-referenced observation matrix $\mathbf{X}_k^{(i)}$ into the new space using the CTP method, the following formula is used:

$$\mathbf{Y}_k^{(i)} = \mathbf{W}_{CTP}^T \mathbf{X}_k^{(i)} \in \mathbb{R}^{2F \times N} \quad (5)$$

The variance of the rows of the obtained $\mathbf{Y}_k^{(i)}$ matrix is typically used as features for the classification problem, although the samples from $\mathbf{Y}_k^{(i)}$ can also be utilized.

D. Extended Common Temporal Pattern (ECTP)

In the ECTP method, time samples containing discriminative information are highlighted [30]. Additionally, trial averages of each class can be employed as prior knowledge to enhance class separation. To implement ECTP, we first construct the temporal filter matrix $\mathbf{W}_1$ by selecting the first F columns from the $\mathbf{W}_{CTP}$ matrix derived from (4). Similarly, we form the $\mathbf{W}_2$ temporal filter matrix by choosing the last F columns from the $\mathbf{W}_{CTP}$ matrix. Next, we calculate the mean data matrices corresponding to the two classes as follows:

$$\bar{\mathbf{X}}_k = \frac{1}{M_k} \sum_{i=1}^{M_k} \mathbf{X}_k^{(i)} \in \mathbb{R}^{T \times N}, \quad k = 1, 2 \quad (6)$$

The two mean data matrices, $\bar{\mathbf{X}}_1$ and $\bar{\mathbf{X}}_2$, will be used as prior information in extracting ECTP features. To obtain all the possible combinations of the filters and the mean data matrices, we compute the filtered average matrices as follows:

$$\bar{\mathbf{Y}}_1 = \mathbf{W}_1^T \bar{\mathbf{X}}_1, \bar{\mathbf{Y}}_2 = \mathbf{W}_2^T \bar{\mathbf{X}}_2 \quad (7)$$

$$\bar{\mathbf{Y}}_3 = \mathbf{W}_1^T \bar{\mathbf{X}}_2, \bar{\mathbf{Y}}_4 = \mathbf{W}_2^T \bar{\mathbf{X}}_1$$

For each trial in the two classes, $\mathbf{X}_k^{(i)}$, we calculate the filtered input matrices as follows:

$$\mathbf{Y}_5 = \mathbf{W}_1^T \mathbf{X}_k^{(i)}, \mathbf{Y}_6 = \mathbf{W}_2^T \mathbf{X}_k^{(i)} \quad (8)$$

$$k = 1, 2 \text{ and } 1 \leq i \leq M_k$$

Subsequently, we create covariance matrices to calculate the similarity of each filtered trial with the mean filtered data matrices:

$$\mathbf{G}_1 = \begin{bmatrix} \mathbf{Y}_5 \\ \mathbf{Y}_6 \end{bmatrix} \begin{bmatrix} \bar{\mathbf{Y}}_1 \\ \bar{\mathbf{Y}}_2 \\ \bar{\mathbf{Y}}_3 \\ \bar{\mathbf{Y}}_4 \end{bmatrix}^T \quad (9)$$

Additionally, we calculate the similarity of each trial with others by creating the following covariance matrix:

$$\mathbf{G}_2 = \begin{bmatrix} \mathbf{Y}_5 \\ \mathbf{Y}_6 \end{bmatrix} \begin{bmatrix} \mathbf{Y}_5 \\ \mathbf{Y}_6 \end{bmatrix}^T \quad (10)$$

All elements within $\mathbf{G}_1$, as well as the upper triangle and diagonal elements of the symmetric matrix $\mathbf{G}_2$, serve as features for classification.

E. Proposed Method

The CTP method is susceptible to noise and overfitting, particularly when training data is limited. To mitigate these issues and highlight time samples with discriminative information, we propose the Regularized Common Temporal Pattern (RCTempP) method, which incorporates a quadratic regularization term into the CTP objective function. The regularization term $\mathbf{Q} \in \mathbb{R}^{T \times T}$ is a diagonal matrix, where each diagonal element q_{ii} is computed based on the average R-squared coefficient across all EEG channels [32]. Specifically, the elements of $\mathbf{Q}$ are defined as:

$$q_{ij} = \begin{cases} 1, & i = j \text{ and } r_i \leq T_{rs} \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where r_i is the R-squared coefficient corresponding to the i -th time sample, and T_{rs} is a predefined threshold. By adding the regularization term $R(\mathbf{w}) = \alpha \mathbf{w}^T \mathbf{Q} \mathbf{w}$, we incorporate prior knowledge derived from the R-squared values into the CTP objective function. The regularization term serves to penalize time samples that are less informative for class discrimination. Specifically, α controls the strength of this penalty: as α increases, the influence of the regularization term becomes stronger. Since $\mathbf{Q}$ assigns a value of 1 to time samples whose R-squared values fall below the threshold T_{rs} , the corresponding weights in $\mathbf{w}$ are discouraged from becoming large. In other words, this mechanism prevents the model from placing too much emphasis on time samples that exhibit low discriminability between classes—i.e., those with low R-squared values (or a biserial correlation value closer to zero)—thus helping to focus learning on the most informative parts of the signal. As the regularization introduces asymmetry into the optimization problem, it must be solved separately for each class in the numerator. As a result, the CTP problem transforms into the following equations:

$$J_{RCTempP_1}(\mathbf{W}_1) = \frac{\mathbf{w}_1^T \mathbf{C}_1 \mathbf{w}_1}{\mathbf{w}_1^T (\mathbf{C}_2 + \alpha \mathbf{Q}) \mathbf{w}_1} \quad (12)$$

$$J_{RCTempP_2}(\mathbf{W}_2) = \frac{\mathbf{w}_2^T \mathbf{C}_2 \mathbf{w}_2}{\mathbf{w}_2^T (\mathbf{C}_1 + \alpha \mathbf{Q}) \mathbf{w}_2} \quad (13)$$

In these equations, the regularized temporal filters are the eigenvectors of the $(\mathbf{C}_2 + \alpha \mathbf{Q})^{-1} \mathbf{C}_1$ and $(\mathbf{C}_1 + \alpha \mathbf{Q})^{-1} \mathbf{C}_2$ matrices, which are then combined into the $\mathbf{W}_{RCTempP}$ matrix:

$$\mathbf{W}_{RCTempP} = [\mathbf{W}_1, \mathbf{W}_2] \quad (14)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ contain the first F eigenvectors of the matrices $(\mathbf{C}_2 + \alpha \mathbf{Q})^{-1} \mathbf{C}_1$ and $(\mathbf{C}_1 + \alpha \mathbf{Q})^{-1} \mathbf{C}_2$, respectively. The resulting temporal filter, $\mathbf{W}_{RCTempP}$, will project each trial into a new space as follows:

$$\mathbf{Y}_k^{(i)} = \mathbf{W}_{RCTempP}^T \mathbf{X}_k^{(i)} \in \mathbb{R}^{2F \times N} \quad (15)$$

where the specific time samples receive higher weights, thereby exerting more influence during the classification step.

TABLE I. TESTED AND SELECTED VALUES FOR EACH HYPERPARAMETER ACROSS DIFFERENT METHODS

Method	Parameters	Tested Values	Selected Values
<i>Time Samples</i>	μ_{Time}	12, 16, 20, 23, 27, 31	16, 20, 23, 27
<i>CTP</i>	μ_{CTP}	7, 10, 13, 17, 20, 23	10, 13, 17, 20
<i>ECTP</i>	μ_{ECTP}	7, 10, 13, 17, 20, 23	10, 13, 17, 20
<i>RCTempP</i>	$\mu_{RCTempP}$ α	4, 7, 10, 13, 16, 19, 22 0.1, 0.2, 0.3, ..., 0.9, 1	7, 10, 13, 16 0.6, 0.8

F. Feature Selection and Classification Procedure

The dataset consists of 5-second trials, with 2 seconds preceding and 3 seconds following the class cue onset (0 s). To evaluate binary classification accuracy, we performed 5-fold cross-validation independently for each subject and for each of the ten possible class pairs. Feature extraction was conducted using a 1-second moving window (16 samples), sliding forward by 1 sample at each step. The first window spanned from -0.5 to $+0.5$ seconds relative to the class cue, and the last window covered $+1.5$ to $+2.5$ seconds. We evaluated four methods: Time Samples, CTP, ECTP, and the proposed RCTempP. For all methods that rely on temporal filters, F was fixed at 8, which includes all available temporal filters. In each fold, temporal filters (where applicable) were derived from the training set and applied to extract features. For the Time Samples method, raw samples were ranked by Fisher score, and the top μ_{Time} were selected. For CTP and ECTP, temporal filters were computed from the training data, applied to obtain matrices Y_{CTP} and Y_{ECTP} , respectively, and the top μ_{CTP} and μ_{ECTP} samples were selected from these matrices based on Fisher score. For RCTempP, temporal filters were derived from the training data using the regularization term α , and applied to obtain $Y_{RCTempP}$. The threshold T_{rs} was defined as the standard deviation of the R-squared coefficients computed on the training set. Then, the top $\mu_{RCTempP}$ samples were again selected using Fisher score. An LDA classifier (MATLAB's *fitcdiscr* function) was trained using the selected training features, and then applied to the test data using the same filters and feature indices. This procedure—feature extraction, training, and testing—was repeated at each window position across the trial. Confusion matrices were recorded for each window for further analysis. To improve robustness, the entire classification pipeline was repeated across 13 rounds per subject and class pair, with a new set of five folds generated in each round.

G. Hyperparameters Selection

To optimize hyperparameters, we conducted a grid search over selected feature counts (μ_{Time} , μ_{CTP} , μ_{ECTP} , $\mu_{RCTempP}$) and the RCTempP regularization coefficient (α). Each configuration was evaluated based on the average peak classification accuracy across 5 training rounds in all subjects. Performance trends were visualized using 2D plots for Time Samples, CTP, and ECTP, and a 3D surface plot for RCTempP (data not shown), helping to identify high-performing settings. Poorly performing combinations were discarded, and the remaining ones were further assessed across 13 classification rounds. Table 1 summarizes the tested and selected values. While a full sensitivity analysis is beyond the scope of this study, we explored a reasonable range of values and selected those yielding the best results.

III. EXPERIMENT AND RESULTS

A. Superior Classification Accuracy of RCTempP Over Time Samples

To compare classification accuracy between RCTempP and Time Samples across subjects and class pairs, the optimal regularization coefficient (α) and number of features (μ) were first determined. For each classification round, the maximum accuracy across all windows was extracted, resulting in 13 peak accuracy values per setting. The average of these 13 values was computed for each α - μ pair to select the optimal configuration per class pair and method. Table 2 reports the average of 13 peak accuracies for each subject and class pair using the optimal parameters for both methods, along with their corresponding unbiased standard deviations (rounded to one decimal). The last row summarizes the mean and standard deviation of the peak accuracies averaged across all subjects for each class pair. To assess statistical significance, a one-tailed paired-sample nonparametric permutation test with $t_{\max}$ family-wise error correction [33] was applied across the 10 class pairs. In Table 2, asterisks denote significant differences with starred cells indicating where either RCTempP or Time Samples significantly outperformed the other. Out of 50 subject-class pair combinations, RCTempP outperformed Time Samples in 16 cases (7 with $p < 0.001$, 6 with $p < 0.01$, and 3 with $p < 0.05$), while Time Samples outperformed RCTempP in only one case ($p < 0.05$). When averaged across subjects (Table 2, last row), RCTempP surpassed Time Samples in 8 out of 10 class pairs (5 with $p < 0.001$, 1 with $p < 0.01$, and 2 with $p < 0.05$). Fig. 1 categorizes the movement classes into three clusters: Supination–Pronation, Palmar–Lateral Grasp, and Hand Open. Analysis of classification results across subjects (Table 2, last row) reveals that class pairs within the same cluster are less discriminable—e.g., SN–PN (63.4%) and PG–LG (64.7%) show lower accuracy, indicating greater inter-class similarity. This suggests that the RCTempP method outperforms the Time Samples method in distinguishing between signals from different clusters, the movement classes that were indeed separable in the 0.3–3 Hz frequency band.

B. Reduced Discriminability of Motor Imagery Tasks Involving the Same Joints

An analysis of the predefined movement clusters (Fig. 1) reveals that classes within each cluster engage similar joints—for example, Supination and Pronation (Cluster 1) involve the wrist and elbow, while Lateral and Palmar Grasps (Cluster 2) involve the wrist and fingers. Classification accuracy is notably lower for within-cluster comparisons: RCTempP achieves 63.4% for SN–PN and 64.7% for PG–LG, while Time Samples yields 63.1% and 64.0%, respectively. In contrast, RCTempP shows significantly higher accuracy in between-cluster classifications, indicating its improved ability to distinguish movements involving different joint groups.

C. Subject-Wise Variability in Class Pair Classification Accuracy

Classification accuracy differs notably between subjects for both the RCTempP and Time Samples methods. For example, in the HO–PG condition, RCTempP achieves 75.9% for subject 6 and 74.8% for subject 8, but drops to 68.7%, 68.6%, and 67.3% for subjects 1, 7, and 3, respectively. Time Samples follows a similar trend, with subject 6 achieving 75.0%, while others range between 66.3% and 69.4%. This

TABLE II. AVERAGE PEAK CLASSIFICATION ACCURACIES AND UNBIASED STANDARD DEVIATIONS FOR RCTempP AND TIME SAMPLES ACROSS DIFFERENT CONDITIONS

Class Pairs		SN & PN	SN & HO	SN & PG	SN & LG	PN & HO	PN & PG	PN & LG	HO & PG	HO & LG	PG & LG
Subjects											
1	<i>RCTempP</i>	61.9 ± 2.2	72.2 ± 2.1	* 70.4 ± 2.4	69.2 ± 2.4	69.8 ± 2.7	67.2 ± 2.4	* 67.5 ± 3.0	68.7 ± 1.6	* 71.2 ± 3.2	** 65.5 ± 2.2
	<i>Time</i>	60.5 ± 2.3	72.8 ± 2.4	66.5 ± 2.8	67.8 ± 2.9	68.7 ± 2.7	63.8 ± 3.4	62.9 ± 2.4	66.3 ± 2.9	66.1 ± 2.4	60.9 ± 2.2
3	<i>RCTempP</i>	61.3 ± 2.5	* 78.8 ± 1.3	** 75.9 ± 2.5	** 77.5 ± 1.7	72.7 ± 2.0	73.3 ± 2.0	* 75.4 ± 2.1	67.3 ± 2.2	65.0 ± 1.5	68.2 ± 2.5
	<i>Time</i>	63.1 ± 2.6	77.2 ± 1.6	70.7 ± 1.8	73.1 ± 1.5	72.6 ± 2.0	72.9 ± 2.2	72.3 ± 2.3	68.9 ± 2.5	65.7 ± 2.9	68.6 ± 1.6
6	<i>RCTempP</i>	63.3 ± 3.5	68.6 ± 3.1	79.2 ± 1.4	* 72.6 ± 2.5	72.8 ± 2.4	77.5 ± 1.6	** 77.4 ± 2.3	75.9 ± 1.8	72.7 ± 2.5	63.3 ± 2.4
	<i>Time</i>	65.0 ± 3.6	68.2 ± 2.3	79.7 ± 1.8	70.2 ± 1.7	71.2 ± 2.2	76.2 ± 1.9	73.7 ± 1.9	75.0 ± 2.6	70.6 ± 2.1	61.8 ± 3.3
7	<i>RCTempP</i>	63.5 ± 1.7	* 67.4 ± 2.0	73.2 ± 3.6	68.2 ± 1.9	72.1 ± 2.4	73.1 ± 1.6	70.6 ± 1.8	68.6 ± 1.9	63.3 ± 1.9	61.3 ± 1.3
	<i>Time</i>	62.3 ± 3.6	65.0 ± 2.1	71.1 ± 2.2	66.5 ± 2.4	71.3 ± 2.3	72.4 ± 2.0	71.3 ± 1.5	68.3 ± 3.2	60.5 ± 2.9	* 62.9 ± 1.7
8	<i>RCTempP</i>	* 67.1 ± 2.4	* 80.4 ± 1.9	* 76.8 ± 1.5	78.1 ± 2.0	80.5 ± 1.8	76.4 ± 1.4	77.6 ± 2.5	* 74.8 ± 1.9	** 79.6 ± 2.3	65.3 ± 2.6
	<i>Time</i>	64.7 ± 2.0	76.7 ± 2.3	74.0 ± 2.3	77.2 ± 2.0	78.9 ± 1.6	75.5 ± 2.7	77.0 ± 2.6	69.4 ± 2.2	71.8 ± 2.4	65.6 ± 2.3
Average	<i>RCTempP</i>	63.4 ± 3.2	** 73.5 ± 5.7	** 75.1 ± 3.9	* 73.1 ± 4.6	* 73.6 ± 4.3	* 73.5 ± 4.0	* 73.7 ± 4.7	* 71.1 ± 4.0	** 70.4 ± 6.3	64.7 ± 3.2
	<i>Time</i>	63.1 ± 3.3	72.0 ± 5.2	72.4 ± 4.9	70.9 ± 4.4	72.5 ± 4.1	72.2 ± 5.1	71.4 ± 5.2	69.6 ± 3.9	67.0 ± 4.7	64.0 ± 3.6

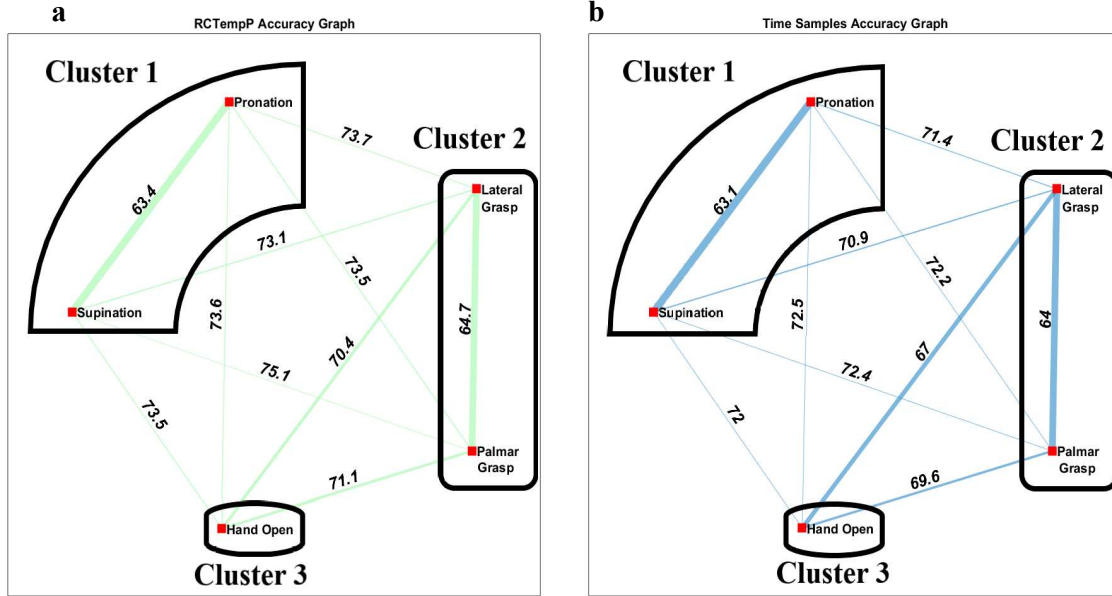
a. Star counts indicate significance levels: * $p < 0.05$, * $p < 0.01$, ** $p < 0.001$.

Fig. 1. Movement clusters defined by the classification accuracy profiles of the RCTempP and Time Samples methods.

inter-subject variability highlights the subject-dependent nature of both methods, which is critical for BCI applications involving motor imagery in individuals with SCI.

D. Comparing the Performance of the Four Different Methods

To evaluate the relative performance of RCTempP, ECTP, and CTP compared to Time Samples, we computed Accuracy, Precision, Recall, and F1 Score from confusion matrices generated for each class pair and method [16]. These metrics were calculated from the sum of confusion matrices across 13 classification rounds per subject and class pair, then aggregated across all five subjects (totaling 65 matrices per method). Table 3 reports the metrics, with the highest values in bold. A one-tailed non-parametric permutation test with $t_{\max}$ family-wise error correction was used to assess statistical significance. The Time Samples method did not outperform

any other method in accuracy for any class pair. In contrast, RCTempP achieved significantly higher accuracy in 8 class pairs (5 with $p < 0.001$, 1 with $p < 0.01$, and 2 with $p < 0.05$). The CTP method outperformed Time Samples in 4 class pairs (2 with $p < 0.001$, 1 with $p < 0.01$, and 1 with $p < 0.05$), and the ECTP method also showed superiority in 4 class pairs (1 with $p < 0.001$, 1 with $p < 0.01$, and 2 with $p < 0.05$). Table 3 also shows that the SN–PN and PG–LG pairs were harder to classify, with CTP and ECTP reaching only 62.7% and 62.0% for SN–PN and 64.8% and 65.8% for PG–LG, respectively. These results are consistent with those of Section III-B, which highlights the challenge of classifying movement classes involving shared joints.

E. Classification Accuracy Across Time

Fig. 2 plots classification accuracy over time for two example class pairs—SN–HO and SN–LG—with accuracy

TABLE III. ACCURACY, PRECISION, RECALL, AND F1 SCORE DERIVED FROM THE CONFUSION MATRICES FOR THE FOUR CLASSIFICATION METHODS

Class Pairs	Metrics	Time	RCTempP	CTP	ECTP	Class Pairs	Metrics	Time	RCTempP	CTP	ECTP
<i>Supination Vs. Pronation</i>	<i>Accuracy</i>	63.1	63.4	62.7	62.0	<i>Pronation Vs. Palmar Grasp</i>	<i>Accuracy</i>	72.3	* 73.5	73.2	72.5
	<i>Precision</i>	63.4	63.8	63.2	62.7		<i>Precision</i>	72.0	73.3	72.9	72.0
	<i>Recall</i>	62.9	62.7	61.9	60.2		<i>Recall</i>	74.1	75.3	75.1	75.0
	<i>F1 score</i>	63.1	63.3	62.6	61.4		<i>F1 score</i>	73.1	74.3	74.0	73.5
<i>Supination Vs. Hand Open</i>	<i>Accuracy</i>	72.0	** 73.6	72.5	71.6	<i>Pronation Vs. Lateral Grasp</i>	<i>Accuracy</i>	71.5	* 73.7	72.0	70.8
	<i>Precision</i>	72.3	74.0	72.6	72.0		<i>Precision</i>	71.2	73.7	71.8	70.9
	<i>Recall</i>	72.3	73.3	73.1	71.6		<i>Recall</i>	72.6	73.9	72.8	71.0
	<i>F1 score</i>	72.3	73.6	72.8	71.8		<i>F1 score</i>	71.9	73.8	72.3	71.0
<i>Supination Vs. Palmar Grasp</i>	<i>Accuracy</i>	72.5	* 75.2	* 73.9	73.0	<i>Hand Open Vs. Palmar Grasp</i>	<i>Accuracy</i>	69.6	* 71.0	* 71.5	* 71.1
	<i>Precision</i>	72.5	75.6	74.3	74.1		<i>Precision</i>	69.5	71.1	71.9	71.3
	<i>Recall</i>	74.1	75.9	74.8	72.1		<i>Recall</i>	71.1	71.9	71.6	71.9
	<i>F1 score</i>	73.3	75.8	74.5	73.1		<i>F1 score</i>	70.3	71.5	71.7	71.6
<i>Supination Vs. Lateral Grasp</i>	<i>Accuracy</i>	71.0	** 73.2	** 73.4	** 73.3	<i>Hand Open Vs. Lateral Grasp</i>	<i>Accuracy</i>	66.8	** 70.2	** 70.1	* 69.6
	<i>Precision</i>	71.4	73.7	74.0	73.9		<i>Precision</i>	66.9	70.8	70.3	70.7
	<i>Recall</i>	70.9	72.9	72.8	72.7		<i>Recall</i>	66.9	68.9	69.8	67.0
	<i>F1 score</i>	71.1	73.3	73.4	73.3		<i>F1 score</i>	66.9	69.8	70.1	68.8
<i>Pronation Vs. Hand Open</i>	<i>Accuracy</i>	72.5	* 73.6	72.6	71.6	<i>Palmar Grasp Vs. Lateral Grasp</i>	<i>Accuracy</i>	64.1	64.7	64.8	* 65.8
	<i>Precision</i>	72.5	73.3	72.3	71.5		<i>Precision</i>	64.1	64.6	64.4	65.8
	<i>Recall</i>	72.8	74.4	73.4	72.2		<i>Recall</i>	62.2	63.3	64.5	64.2
	<i>F1 score</i>	72.7	73.8	72.9	71.8		<i>F1 score</i>	63.2	64.0	64.4	65.0

b. Star counts indicate significance levels: * $p < 0.05$, * $p < 0.01$, * $p < 0.001$. Bold entries indicate the maximum value in each row.

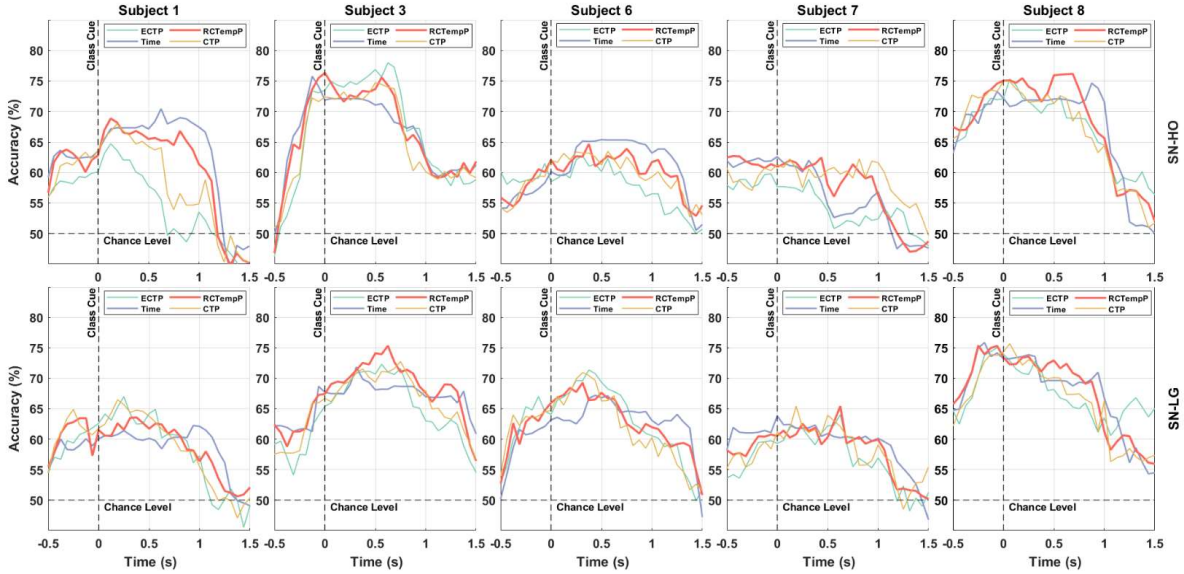


Fig. 2. Classification accuracy over time for each subject in two example class pairs—Supination–Hand Open (SN–HO) and Supination–Lateral Grasp (SN–LG)—using four different methods.

shown at each window's start. RCTempP generally outperforms Time Samples for SN-HO in subjects 3, 7, and 8, and for SN-LG in subject 3. However, the RCTempP method shows no clear advantage for SN-HO in subjects 1 and 6. These conditions, marked with stars in Table 2, indicate that windows with superior peak accuracy in RCTempP also show consistently better temporal performance. Fig. 2 also highlights inter-subject variability in performance, reinforcing

the subject-dependency observed in Section III-C.

IV. DISCUSSION

This study proposes RCTempP, a time-domain feature extractor that leverages R-squared-based regularization to improve EEG classification of imagined arm and hand movements in SCI patients. We tested RCTempP on five subjects from [15]. Each subject imagined five distinct hand

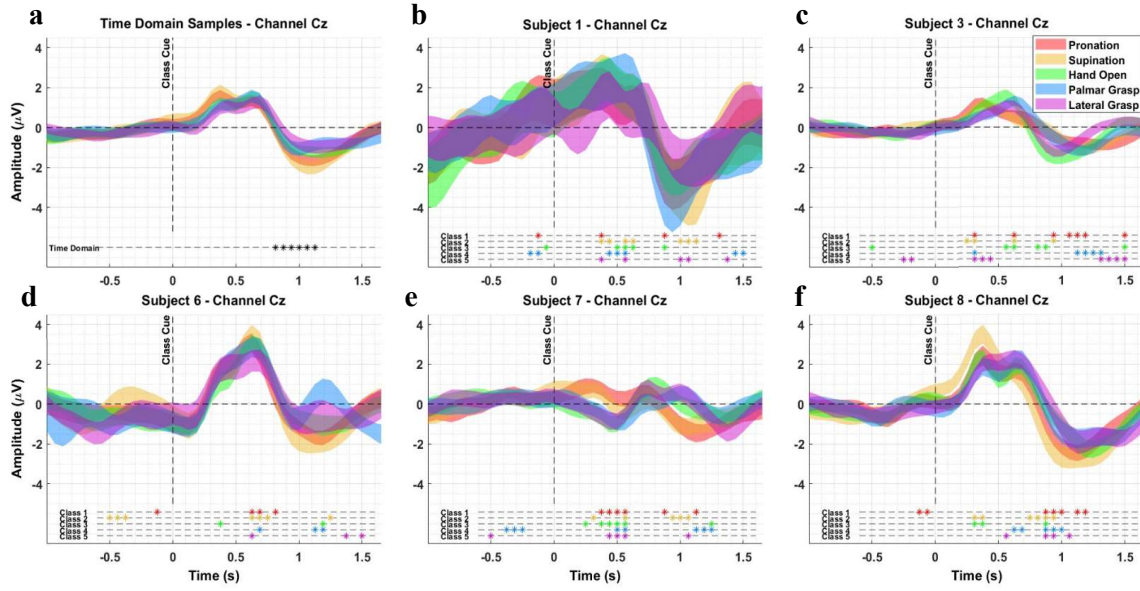


Fig. 3. Movement-related cortical potentials (MRCPs) averaged across all subjects (a) and individually for each subject (b–f), shown with 95% confidence intervals across different movement classes.

movements, and on average RCTempP outperformed raw time samples in eight of the ten tested conditions. In contrast, CTP and ECTP exceeded Time Samples in fewer than half of the conditions. Our analysis was limited to the delta band (0.3–3 Hz). This focus may have constrained ECTP, which captures energy-related features. Additionally, movements that share the same joints were harder to classify, a finding that matches the findings of [22]. We also found that accuracy is generally lower before the cue onset than after it, which confirms that the delta band holds discriminative information after the cue, as reported in [27]. To identify informative timepoints on channel Cz, we first averaged trials per subject s and class c :

$$\bar{x}_{sc}(t) = \frac{1}{M_{sc}} \sum_{i=1}^{M_{sc}} x_{sc,i}(t) \quad (16)$$

$$s = 1, 3, 6, 7, 8 \text{ and } c = 1, 2, 3, 4, 5$$

where $x_{sc,i}(t) \in \mathbb{R}^{T \times 1}$ denotes the preprocessed signal on channel Cz for trial i at time t , and M_{sc} is the total number of trials for that subject–class combination. Next, for each of the ten class pairs (c, c') , we computed:

$$d_{s,cc'}(t) = |\bar{x}_{sc}(t) - \bar{x}_{sc'}(t)| \quad (17)$$

$$s = 1, 3, 6, 7, 8 \text{ and } c, c' = 1, 2, 3, 4, 5 \text{ and } c \neq c'$$

to form the alternate distribution of 50 values per time sample. We then permuted trial labels for each subject to create a null distribution. Because both distributions are nonnegative and non-Gaussian, we used a nonparametric permutation test with $t_{\max}$ family-wise error correction over 32 samples. A time sample was deemed significant if the alternate distribution exceeded the null distribution at $p < 0.05$. These time samples are marked with stars in Fig. 3(a). The class-averaged Cz waveforms (with 95% confidence intervals using a $1.96 \times \sigma$ multiplier) are overlaid in different colors. The interval $[0.813, 1.125]$ s—encompassing the MRCP negative peak—emerges as significant, indicating this peak's discriminative value

during hand imagery and reaffirming the post-cue delta band's importance [22],[34]. We then tested RCTempP's temporal filters by solving a generalized eigenvalue problem for each class pair (c, c') in two windows: $[-0.5, 0.5]$ s and $[0.5, 1.5]$ s. This produced a filter matrix $\mathbf{W}_{cc'}(t)$ whose columns $\mathbf{w}_{cc',r}(t)$ are temporal filters. The first filter $\mathbf{w}_{cc',1}(t)$ maximizes the energy of class c over class c' , while the 16th filter $\mathbf{w}_{cc',16}(t)$ does the opposite. For each subject and window, we compute 65 matrices $\mathbf{W}_{cc'}(t)$ per class pair. We then define the c -vs-others filter as follows:

$$\mathbf{w}_{c,others}(t) = \begin{cases} |\mathbf{w}_{cc',1}(t)|, & c > c' \\ |\mathbf{w}_{cc',16}(t)|, & c < c' \end{cases} \quad (18)$$

$$c, c' = 1, 2, 3, 4, 5 \text{ and } c \neq c'$$

and aggregated 260 filters per window (5 folds $\times$ 13 rounds $\times$ all classes except c). We compared their absolute values to those from permuted labels using the same permutation test. The results from the significant temporal filters (Fig. 3(b–f)) indicate that discriminative information is concentrated around both the positive and negative peaks of the MRCPs following cue onset, consistent with the findings of [15]. However, the significant time samples are primarily located around the negative peak of the MRCPs (Fig. 3(a)). Moreover, Subject 1 consistently showed lower accuracy than Subject 8 (Table 2). This difference aligns with the distinct Cz waveforms in Fig. 3, as Subject 1's MRCPs are more dispersed, whereas those of Subject 8 are more discriminable. These results underscore the subject-dependent nature of these algorithms. In summary, RCTempP's temporal filters capture discriminative information that raw time samples often overlook by emphasizing key MRCP peaks following cue onset and achieving higher classification accuracy. This demonstrates the method's potential to improve BCI performance in motor rehabilitation for SCI patients. The observed variability in MRCP patterns across subjects also underscores the need for further investigation into the neural factors underlying these individual differences.

V. CONCLUSION

This study presents a novel feature extraction method, RCTempP, designed to improve EEG-based decoding of arm and hand movements for BCI applications in SCI patients. RCTempP significantly outperforms the traditional Time Samples method by extracting discriminative temporal features, particularly after cue onset, thereby enhancing classification accuracy. Our findings support the importance of time-domain features in distinguishing hand movements, aligning with prior research on joint involvement, subject-specific variability, and the discriminative role of the delta band. Despite promising results, limitations remain. The small sample size and limited class diversity may affect generalizability. While this study focused on motor imagery in SCI, RCTempP is broadly applicable to other EEG-based BCI tasks. Future research should evaluate its performance on larger datasets, examine its neurophysiological relevance, explore hyperparameter sensitivity, and assess real-time usability. Overall, RCTempP offers a promising avenue for improving motor rehabilitation and advancing BCI systems.

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